

RFID-Based Unified Citizen Service Management System

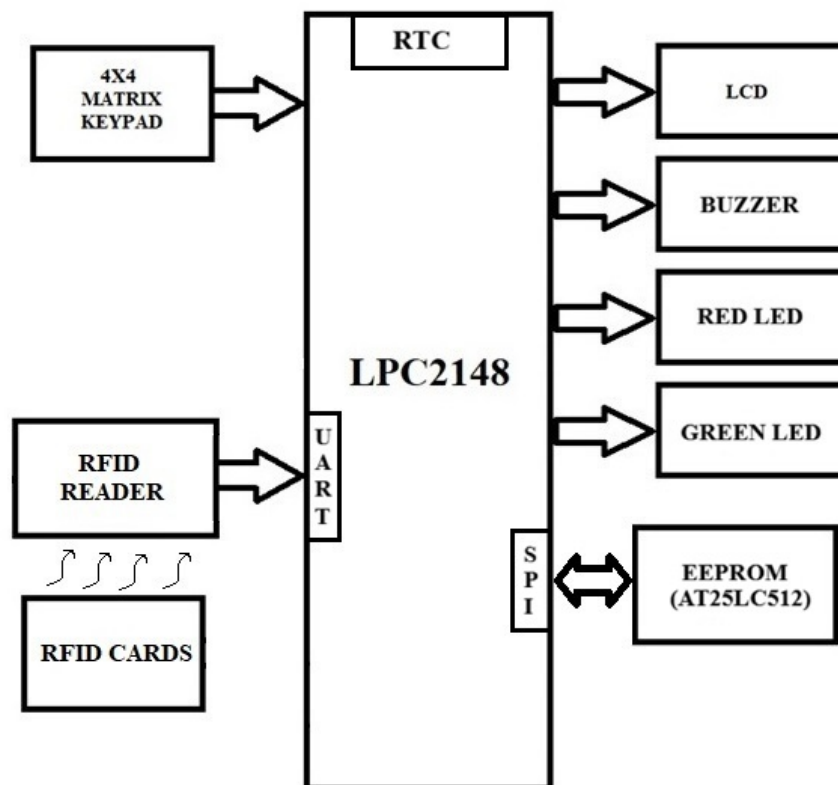
AIM:

To design and develop an RFID-Based Unified Citizen Service Management System that enables secure user authentication and provides multiple citizen services such as digital identity verification, banking operations, electronic voting, and driving license validation through a single RFID-based platform.

INSIGHT:

- Good knowledge of C-Programming
- Knowledge of Embedded-C programming.
- Thorough with the LPC2148 Architecture, General purpose I/O Interface.
- Understanding of RFID usage.
- Understanding of SPI protocol.
- Understanding of LCD and KEYPAD.

BLOCK DIAGRAM:



Note: Apply 3.3V as an operating voltage to SPI based EEPROM (AT25LC512)

REQUIREMENTS:**HARDWARE REQUIREMENTS:**

- LPC 2148
- RFID Reader
- RFID cards
- 20x4 LCD
- 4x4 Matrix keypad
- AT25LC512
- LED'S
- BUZZER
- USB-UART Converter

SOFTWARE REQUIREMENTS:

- EMBEDDED C – PROGRAMMING
- KEIL-C COMPILER
- FLASH MAGIC

SEQUENCE TO BE FOLLOWED FOR IMPLEMENTATION:

- Create New Folder in your server/Laptop/PC and save that folder with your project name
- Individually can check each module.
- First check lcd to display character constant, string constant and integer constant.
- Next check keypad peripheral by displaying key values on LCD and design one function for password. This function will return true or false based on the password entry from us.
- Next write n bytes into EEPROM and read that n number of bytes from EEPROM and display on LCD.
Note: Use BYTE WRITE and BYTE READ functions
- Next check the UART peripheral to transmit character constant, string constant and integer constant and receive string constant using UART interrupts (download UART interrupts code from LMS).
- Next connect RFID reader to LPC2148 microcontroller then write logic for reading the RFID card information and display on LCD.

Selected RFID readers will give 10 bytes of data for each card. First character is start of the text (0x02), 8 bytes of card number and last character is end of the text (0x03).

If the card number is “12345678” then output of the RFID reader is 0x02, 0x31, 0x32, 0x33, 0x34, 0x35, 0x36, 0x37, 0x38, 0x03

Note: need to extract the only card number from this data

- After checking all individual interfacing, then follow the below-mentioned steps to complete the project.
- Display Project Name on LCD for some time.
- Then display “Waiting for card” String in 1st line on LCD
- Once the RFID card is placed in front of RFID reader then reader is sending the card number through serial communication with the help of 9600 baud rate.
- After reading the card number, display card number equivalent username on lcd if card is valid otherwise display “invalid card” on LCD. If card is valid green led will blink otherwise red led will blink along with buzzer.
- If card belongs to user card, then display below mentioned menu on LCD.

1. PAN CARD
2. ATM CARD
3. VOTING CARD
4. DRIVING LICENSE
5. EXIT

- From that menu, user must select one option based on his/her requirement.
- If any one of the options is selected among first three, then need to enter the password to access that operation. If entered password is correct, then required operation need to perform.
- If option 1 is selected, then display the user’s name, date of birth and pan card number on LCD for some time then control goes back to menu.
- If option 2 is selected, display the below mentioned menu on LCD.
 1. Bal Enquiry
 2. Withdrawn
 3. Deposit
 4. exit
- Based on the selected option, need to perform the operation. If bal enquiry is selected, then read the user account balance from EEPROM from fixed location and display on LCD.

Note: for balance, need to take two memory locations (maximum balance should be 65535)

- If withdrawn option is selected, then check if the sufficient balance is available or not. If available, then only do the process otherwise don't process it.
- Note: need to maintain the 500 rupees minimum balance.
- If deposit option is selected, then add the money to account. This option is just for demonstration purposes only.
- Note: need to add required test cases according to real time process.
- If voting option is selected, first check the voting status of user. If voting is not done, then display the below-mentioned menu to allow user for voting.

1. PARTY1 – SYMBOL
2. PARTY2 - SYMBOL
3. PARTY3 - SYMBOL
4. PARTY4 – SYMBOL
5. EXIT

Note: users must select any one of the options. And user defined symbols are designed with the help of CGRAM concept.

For reference use the following keypad array.... (not compulsory, you can use as per your interest)

'1'	'2'	'3'	'4'
'5'	'6'	'7'	'8'
'9'	'0'	'*'	'#'
'!'	'@'	'\$'	'&'

* - delete

- enter

- Based on the voter interest he/she must select the one party which he/she has to vote for.
- Here EEPROM is used to save the voting status of the user.
- If user is selected as the 4th option, then display the user driving license information on LCD.
- Driving license information should be username, Class of vehicle (two/four-wheeler),

license validity, user address in short cut manner.

- To check the license validity, first read the current RTC information from the on-chip RTC and compare it with the license last date. If it is valid then display the “valid license” on LCD or display “Invalid license. Please renew your license” on LCD.

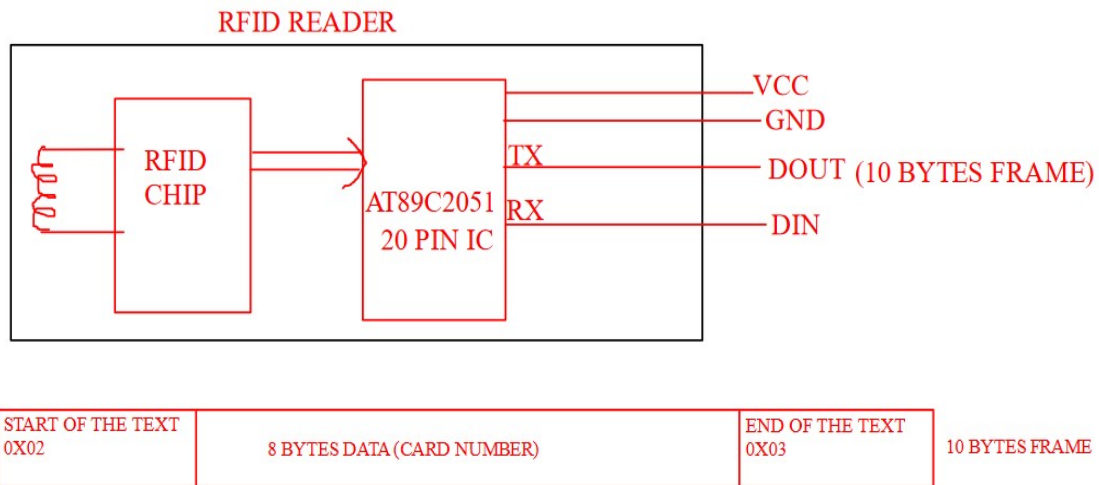
Note: last date of driving license information needs to store in the code memory of LPC2148.

- If 5th option is selected, then control goes back to menu.
- The above process for user card. If officer card is placed, then it is used for resetting the voting machine. After resetting the voting status, again user can utilize his/her vote.
- For the 4th option, RTC information is very important. Due to any reason RTC is not showing the correct time then time editing is required. To edit the rtc information, need to generate the external interrupt by pressing the switch which is connected to interrupt pin. Menu based rtc edit need to do.
- If you get this output, then your project is completed.

***** ALL THE BEST *****

RFID Reader:

RFID Reader will send 10 bytes of data once card is placed nearer to the RFID Reader. Once check the below images to get some idea on RFID Reader block diagram. Refer the supported data sheets to get some more knowledge.



For example, card number is 12345678, then output of the RFID Reader is
0x02 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x03 (hex format)